TVM vs Polly SCoP

```
for t in T.serial(2048):
  for h in T.serial(16):
    for k in T.serial(2048):
      for rd in T.serial(512):
        with T.block("logits_n"):
          v_t = T.axis.spatial(2048, t)
          v_h = T.axis.spatial(16, h)
          v_k = T.axis.spatial(2048, k)
          v_rd = T.axis.reduce(512, rd)
          T.reads(
            q_nope[v_t:v_t+1, v_h:v_h+1, v_rd:v_rd+1],
            Kc_gather[v_t:v_t+1, v_k:v_k+1, v_rd:v_rd+1]
          )
          T.writes(
            logits_n[v_t:v_t+1, v_h:v_h+1, v_k:v_k+1]
          )
          logits_n[v_t, v_h, v_k] = (
            logits_n[v_t, v_h, v_k]
            + T.Cast("float32", q_nope[v_t, v_h, v_rd])
            * T.Cast("float32", Kc_gather[v_t, v_k, v_rd])
          )
```

```
Scop: {
  Statements:
    S_red[t, h, k, rd]:
      Domain:
        { S_red[t, h, k, rd] :
            0 <= t  < 2048 and
            0 <= h  < 16   and
            0 <= k  < 2048 and
            0 <= rd < 512 }
      ReadAccess:
        q_nope[t, h, rd]
        Kc_gather[t, k, rd]
        logits_n[t, h, k]
      WriteAccess (reduction):
        logits_n[t, h, k]
      Body:
        logits_n[t, h, k] =
            logits_n[t, h, k]
            + q_nope[t, h, rd]
            * Kc_gather[t, k, rd]
  Schedule:
     S_red[t, h, k, rd]
     -> (t, h, k, 1, rd)
}
```

TVM vs Polly SCoP

Loops

```
for t in T.serial(2048):
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    for k in T.serial(2048):
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          T.reads(
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            Kc_gather[v_t:v_t+1, v_k:v_k+1, v_rd:v_rd+1]
          )
          T.writes(
            logits_n[v_t:v_t+1, v_h:v_h+1, v_k:v_k+1]
          )
          logits_n[v_t, v_h, v_k] = (
            logits_n[v_t, v_h, v_k]
            + T.Cast("float32", q_nope[v_t, v_h, v_rd])
            * T.Cast("float32", Kc_gather[v_t, v_k, v_rd])
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```

```
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  Statements:
    S_red[t, h, k, rd]:
      Domain:
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            0 <= h  < 16   and
            0 <= k  < 2048 and
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        q_nope[t, h, rd]
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        logits_n[t, h, k] =
            logits_n[t, h, k]
            + q_nope[t, h, rd]
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```

Schedule

(more info in AST)

```
Schedule:
   S_red[t, h, k, rd]
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TVM vs Polly SCoP

Loops

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  for h in T.serial(16):
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```

Iter vars

```
v_t = T.axis.spatial(2048, t)
v_h = T.axis.spatial(16, h)
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v_rd = T.axis.reduce(512, rd)
T.reads(
  q_nope[v_t:v_t+1, v_h:v_h+1, v_rd:v_rd+1],
  Kc_gather[v_t:v_t+1, v_k:v_k+1, v_rd:v_rd+1]
)
T.writes(
  logits_n[v_t:v_t+1, v_h:v_h+1, v_k:v_k+1]
)
logits_n[v_t, v_h, v_k] = (
  logits_n[v_t, v_h, v_k]
  + T.Cast("float32", q_nope[v_t, v_h, v_rd])
  * T.Cast("float32", Kc_gather[v_t, v_k, v_rd])
)
```

Domain

```
Scop: {
  Statements:
    S_red[t, h, k, rd]:
      Domain:
        { S_red[t, h, k, rd] :
            0 <= t  < 2048 and
            0 <= h  < 16   and
            0 <= k  < 2048 and
            0 <= rd < 512 }
      ReadAccess:
        q_nope[t, h, rd]
        Kc_gather[t, k, rd]
        logits_n[t, h, k]
      WriteAccess (reduction):
        logits_n[t, h, k]
      Body:
        logits_n[t, h, k] =
            logits_n[t, h, k]
            + q_nope[t, h, rd]
            * Kc_gather[t, k, rd]
```

Schedule

(more info in AST)

```
      Schedule:
         S_red[t, h, k, rd]
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}
```

TVM vs Polly SCoP

Loops

```
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  for h in T.serial(16):
    for k in T.serial(2048):
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```
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v_k = T.axis.spatial(2048, k)
v_rd = T.axis.reduce(512, rd)
T.reads(
  q_nope[v_t:v_t+1, v_h:v_h+1, v_rd:v_rd+1],
  Kc_gather[v_t:v_t+1, v_k:v_k+1, v_rd:v_rd+1]
)
T.writes(
  logits_n[v_t:v_t+1, v_h:v_h+1, v_k:v_k+1]
)
```

Stmts

```
logits_n[v_t, v_h, v_k] = (
  logits_n[v_t, v_h, v_k]
  + T.Cast("float32", q_nope[v_t, v_h, v_rd])
  * T.Cast("float32", Kc_gather[v_t, v_k, v_rd])
)
```

```
Scop: {
  Statements:
    S_red[t, h, k, rd]:
```

Domain

```
Domain:
  { S_red[t, h, k, rd] :
      0 <= t  < 2048 and
      0 <= h  < 16   and
      0 <= k  < 2048 and
      0 <= rd < 512 }
ReadAccess:
  q_nope[t, h, rd]
  Kc_gather[t, k, rd]
  logits_n[t, h, k]
WriteAccess (reduction):
  logits_n[t, h, k]
```

Stmts

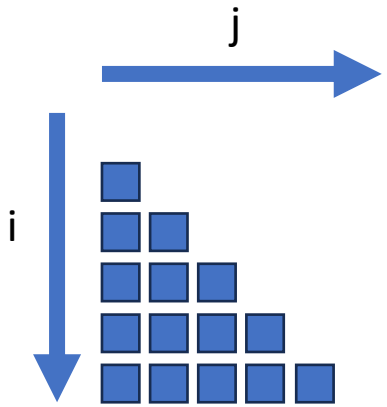
```
Body:
  logits_n[t, h, k] =
      logits_n[t, h, k]
      + q_nope[t, h, rd]
      * Kc_gather[t, k, rd]
```

Schedule

(more info in AST)

```
Schedule:
   S_red[t, h, k, rd]
   -> (t, h, k, 1, rd)
}
```

```
for (int i = 0; i < 6; i++) {
    for (int j = 0; j < i; j++) {
        a[i][j]++;
    }
}
```

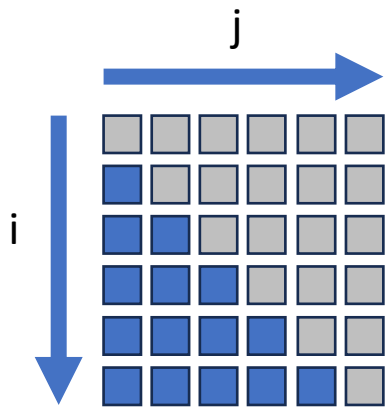


TVM (and the GPU) doesn't like this ☹

```
for (int i = 0; i < 6; i++) {
    for (int j = 0; j < 6; j++) {
        if (j < i)
            a[i][j]++;
    }
}
```

Use if statement masking

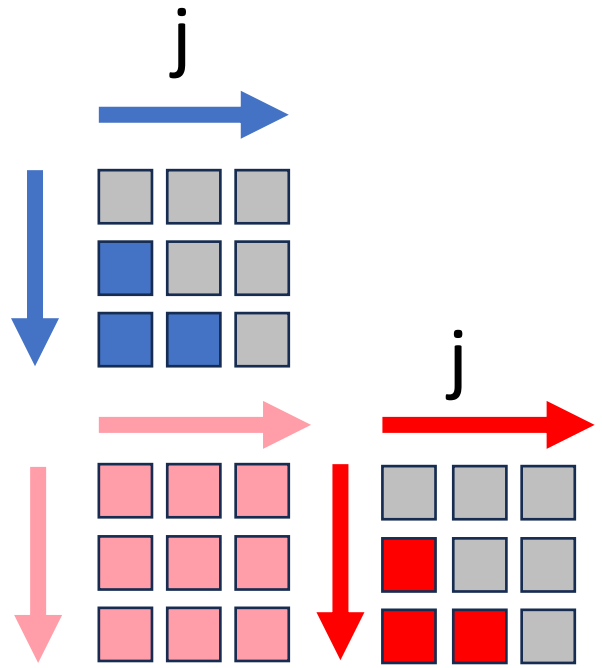
Find an enclosing rectangle bound



Non-rectangular bounds can be converted into rectangular ones by adding if statement masking

This results in less warp divergence on GPUs and is supported by TVM

```
for i = 0 to 3 {
    for j = 0 to i {
        if (j < i)
            a[i][j]++;
    }
}
for i = 3 to 6 {
    for j = 0 to 3 {
        a[i][j]++;
    }
}
```




```
for i = 3 to 6 {
    for j = 3 to i {
        if (j < i)
            a[i][j]++;
    }
}
```

Another strategy: create multiple rectangles (minimize masked iterations)

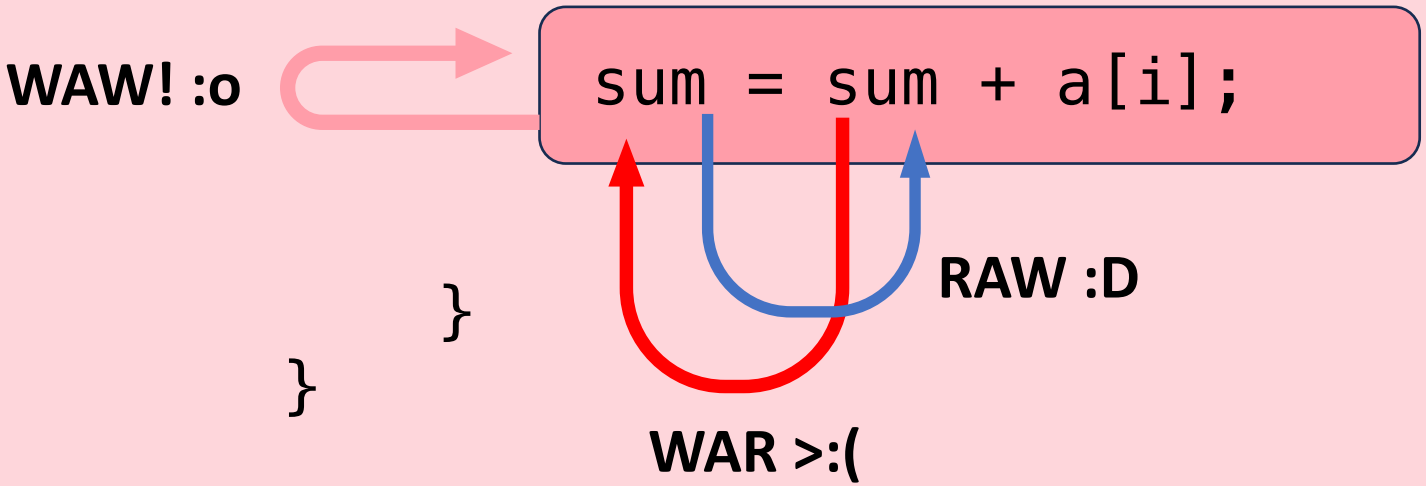
Downsides:

- Increases program size and auto-tuning search space
- Profitability is questionable
- Hard to choose a good set of bounding boxes

Let us have a look at reductions

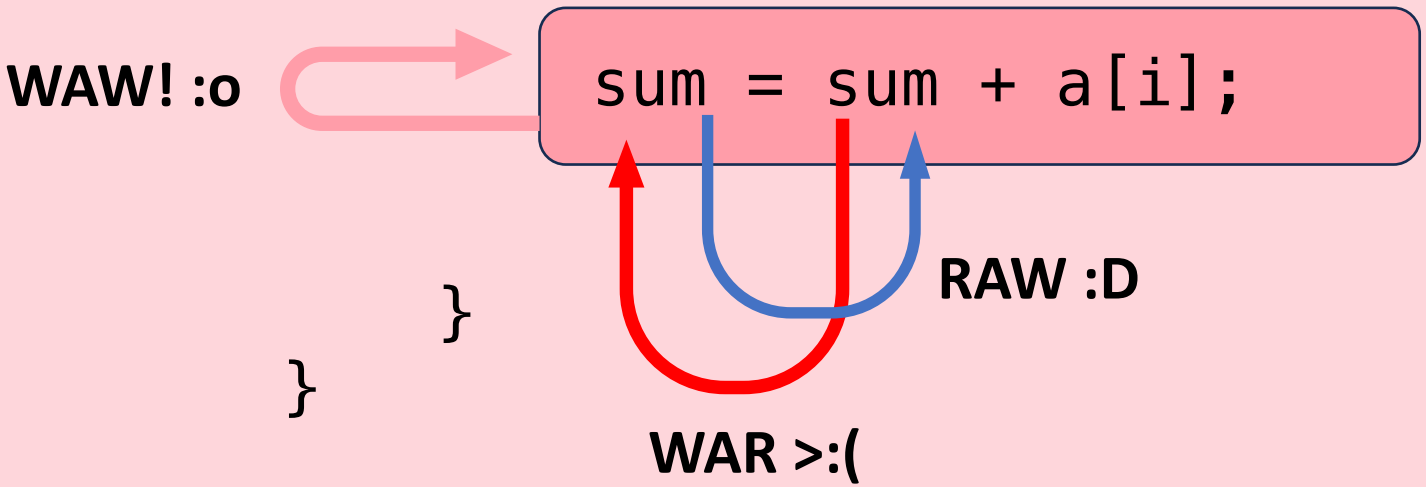
(Since apparently we didn't see enough dependencies yesterday)

```
void reduction(int *a, int n) {
    for (int i = 1; i < n; i++) {
        sum = sum + a[i];
    }
}
```



Polly gives us reduction detection for “free”! *Waw!*

```
void reduction(int *a, int n) {
    for (int i = 1; i < n; i++) {
        sum = sum + a[i];
    }
}
```



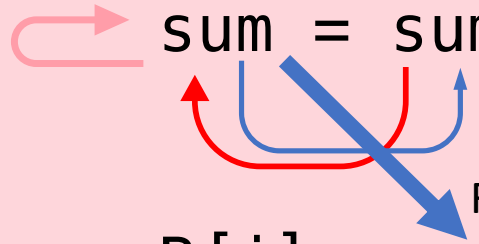
However…what if we add another statement?

```
void reduction(int *a, int n) {
    for (int i = 1; i < n; i++) {

        sum = sum + a[i];

        B[i] = sum;

    }
}
```



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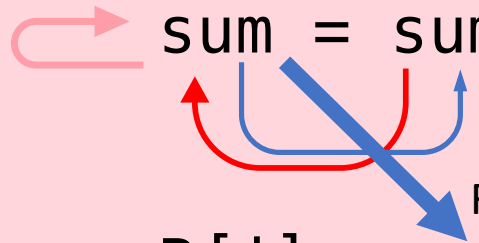

In this case, the leaking partial sums make this a scan.

TVM needs to know about scans, but Polly does not recognize them.

Takeaways

- Domain specific languages can be used to accelerate some types of kernels
- However, it can be hard to write code in them
- Hence, we extract and translate general purpose program parts into DSLs

“LaTeaVM”